**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING**
**NORTH CAROLINA STATE UNIVERSITY**

**CHE 450** **Homework Set 2**

**Problem 1 (90%)** Suppose you are scheduling workflow at a batch processing facility which produces three products (A, B and C) using the same main pieces of equipment (mixer, reactor, separator). The products require different processing times in each piece of equipment (in hours, h), as shown here:

| Product | Mixer | Reactor | Separator |
|---|---|---|---|
| A | 2.0 h | 5.0 h | 4.0 h |
| B | 3.0 h | 4.0 h | 3.5 h |
| C | 1.0 h | 3.0 h | 4.5 h |

It is specified that the facility produce the same number of batches of each product. Assume you have unlimited intermediate storage (i.e. you can store any of the products at any point in their production while waiting for the next piece of equipment to become available if needed). *Note:* you can use Excel in producing Gantt charts for the following problems if you wish, but all materials must be submitted through Gradescope. Do not submit pictures of entire Excel spreadsheets, as this risks deductions for poor submission clarity / visibility, and please also do not email your Excel files to the instructor or TAs/graders. Instead, you may need to "wrap" your tables on the page of a Word document (i.e. paste as much as you can on one line of the page, then paste as much of the rest of the table on the next line, etc.) for each of the subparts of the problem to ensure your PDF submission is legible. You should be able to fit each 50 hour Gantt chart on a single page by orienting the page in landscape format and having 3 sections of chart.
**(a, 30%)** *(i, 20%)* Produce a Gantt chart which shows the first 50 hours of production using single-product campaigns (put another way, you would start production of A, and once its production was entirely complete you would start production of B, etc.). *(ii, 10%)* How many total batches of product (A, B and C combined) can you make in 500 hr using this strategy?
**(b, 30%)** *(i, 20%)* Produce a Gantt chart which shows the first 50 hours of multiproduct production using the sequence ABCABC... (put another way, you would start by mixing Product A, and once its time in the mixer was complete you would move Product A to the reactor while simultaneously starting Product B in the mixer, etc.). *(ii, 10%)* How many total batches of product (A, B and C combined) can you make in 500 hr using this strategy?
**(c, 30%)** *(i, 20%)* Produce a Gantt chart which shows the first 50 hours of multiproduct production using the sequence CBACBA... (put another way, you would start by mixing Product C, and once its time in the mixer was complete you would move Product C to the reactor while simultaneously starting Product B in the mixer, etc.). *(ii, 10%)* How many total batches of product (A, B and C combined) can you make in 500 hr using this strategy?

**Problem 2 (10%)** Suppose you are working as part of an engineering team as you consider the following scenarios. Points will not be given for answers that are unrealistic, unprofessional or otherwise not valid.
**(a, 5%)** It is Friday evening and you have not caught up on you work that must be completed by Monday morning. Normally you would complete your work over the weekend, but you can’t on this particular weekend because you will be driving to an out-of-state wedding where many of your

closest friends will be in attendance. How would you handle this situation with your team (e.g. would you plan on working around the wedding? What might you communicate to your team?).
**(b, 5%)** Your team is determining which of three approaches to use toward solving a problem. *(i, 2%)* Do you think allowing team members to vote anonymously is the best way to choose one of these three paths moving forward? Explain. *(ii, 3%)* What alternative method of resolving the conflict (as opposed to voting) would you suggest? Explain why you think your proposed alternative resolution method is a good idea.